

Raga Mouni Batchu

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SUMMARY

Software Engineer with 6+ years automating cloud-native infrastructure and CI/CD delivery for AI-powered production systems. Hands-on experience with Infrastructure as Code (Terraform), containerized platforms (Docker, Kubernetes), and REST API-driven automation frameworks that reduce manual operational work. Built observability and monitoring practices with Prometheus and Grafana to improve release quality and platform stability. Served as Scrum Master for 3 years, embedding delivery discipline, documentation, and coaching into agile engineering teams.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C++, Perl

AI/ML Frameworks & Libraries: PyTorch, Hugging Face Transformers, LangChain, Vertex AI, PySpark

AI/ML Techniques: Supervised Learning, Instruction-Based Fine-Tuning, Retrieval-Augmented Generation (RAG), GraphRAG, AI Agent Development, LLM Evaluation, Prompt Engineering

Databases: PostgreSQL, MongoDB, DynamoDB, SQL, NoSQL, Snowflake

Backend & APIs: Spring Boot, Node.js, GraphQL, Hasura, RESTful API, FastAPI, OAuth 2.0, SAML

Cloud Platforms: Google Cloud (Cloud Run, Vertex AI), Firebase; AWS (S3, SQS, RDS, SSM, EC2, ECS, CloudFront, CloudWatch, Lambda, EventBridge, Bedrock)

Infrastructure & DevOps: Docker, Kubernetes, AWS CDK, Terraform, Terragrunt, Jenkins, CI/CD, Git

Data & Messaging: Apache Kafka, Informatica, RabbitMQ

Monitoring, Logging & API Tools: Grafana, Prometheus, Tableau, Graylog, Postman, cURL

EDUCATION

West Chester University of Pennsylvania

01/2025 - 05/2026

Master of Science in Computer Science (GPA: 4/4)

Shri Vishnu Engineering College for Women

06/2015 - 05/2019

Bachelor of Technology in Computer Science (GPA: 9/10)

RELATED EXPERIENCE

PennText Education, Pennsylvania, United States of America

Lead AI Engineer

05/2026 – Current

- Optimized Gemini prompts and inference with stable prompt prefixes for implicit KV caching, concise prompts and tool-call arguments to reduce decoding overhead, and parallel execution of independent tool calls to reduce latency and token usage.
- Built LAYERS' (PennText's memory platform) multimodal extraction pipeline with Gemini on Vertex AI; evaluated quality, latency, and cost, and shipped a 3-tier fallback cascade that cut cost ~68% within a 150s SLA.
- Built CoursePackPlus with a multi-turn AI interview and RAG enrichment pipeline to generate publishable coursepacks, with GitHub-webhook sync into LAYERS.
- Deployed containerized AI services on Google Cloud Run, integrating Firebase for application services and Vertex AI for Gemini inference across PennText workflows.
- Delivered CRM and lead-pipeline automation using serverless functions and Gmail API, unifying contacts, interactions, and sales data with automated lead notifications.

iPipeline, Pennsylvania, United States of America

AI Engineer Intern

01/2026 – 05/2026

- Provisioned and managed AWS infrastructure as code with Terraform for an agentic document-digitization service, defining reusable modules for compute, storage, and vector-search resources to stand up environments consistently. Set up CloudWatch alarms and logging for infrastructure health monitoring, integrated into the Terraform-provisioned environment.
- Built the core workflow using LLMs to interpret and map paper-based insurance forms to digital templates through vector comparison and automated tool execution on AWS Bedrock.
- Built MCP server integrations and REST APIs that ingest user-provided insurance data, validate and transform it into structured representations, and automatically drive downstream workflows including form-filling and quote retrieval, cutting manual data-entry steps across the pipeline.

athenahealth, Bengaluru, India

Senior Software Engineer

06/2023 – 08/2024

- Led the migration of call-related features from a legacy system to microservices, adding new functionalities and saving \$732k annually by reducing manual client team workloads.
- Designed and implemented CI/CD pipelines (Jenkins) for the microservices migration, automating build, test, and deployment stages and cutting release cycle time from 3 days to 1 day.
- Built a comprehensive monitoring system for messaging microservices using Prometheus for metrics collection and Grafana dashboards, giving the team real-time operational visibility and faster detection across services.
- Architected the microservices around async, event-driven model with nginx for load balancing across services, and implemented OAuth-based login authentication with SAML SSO integration for enterprise client access.
- Standardized containerized build and deployment workflows (Docker) across the messaging microservices, enabling consistent environment parity from dev through production.

Software Engineer

06/2021 – 05/2023

- Developed an email blacklist feature that eliminated hard bounces, significantly improving KPR.
- Extended SMS services to support Spanish, increasing accessibility for the US-based user base.
- Integrated automated testing gates into the deployment pipeline for the Two-Way SMS service, catching failures before they reached the 100,000-message-per-day production workload.

Associate Software Engineer

06/2019 – 05/2021

- Implemented a last-minute reminder system for patient appointments, reducing no-show rates by 18%.
- Resolved high-priority bugs impacting production systems, improving product stability and client satisfaction.

TiVo, Bengaluru, India

06/2018 – 07/2019

Software Engineer Intern

- Enhanced the TV on-screen guide search experience, improving usability and content discoverability.
- Developed features to preserve user state during navigation, ensuring a seamless transition when returning to previous screens.

RESEARCH & PUBLICATIONS

Bidirectional Translation of ASL and English using CNN and Transformers

Published

- Built a bidirectional ASL-English translation system combining a CNN-BiLSTM sign-to-gloss model, Transformer-based gloss-to-sentence models, and a gloss-to-sign animation pipeline.

A Survey on Data Visualization Techniques for International Trade

Published

ASL recognition and game-based interaction for Deaf Learners

Published

- Developed an ASL learning game with a custom backtracking-based puzzle generator and a spatial-temporal Transformer for gloss recognition, achieving 89% Top 5 accuracy on WL-ASL dataset.

KG-RAG for Tox21: Explainable Drug Toxicity Prediction

Submitted

- Developing a knowledge graph-based RAG application on the toxicity domain, integrating Tox21, PubChem, DSSTox, ChemBL, UniProt and Reactome data products.

PROJECTS

- **Novel Generator using LLM Agentic Framework and RAG Pipeline:** Designed a tool that takes user-provided information, generates novel chapters using an LLM Agentic framework and a RAG pipeline, extracts metadata from the generated content, and stores it in a vector database for use in subsequent chapter generations.
- **Fine-Tuning Flan-T5 for Entity Extraction:** Fine-tuned the Flan-T5 model on a custom instruction-based entity extraction dataset using LoRA, improving extraction accuracy by approximately 25% compared to the original Flan-T5 model. [GitHub](#)
- **Organ Donor & Recipient Matching DApp:** Developed a decentralized application using Ethereum Solidity and Web 3.0 to match organ donors with recipients.

CERTIFICATIONS & ACHIEVEMENTS

- Certification in “Generative AI with Large Language Models” - Coursera
- Awarded Starlight Award for 4 releases consecutively at athenahealth]
- Winner, ConsenSys India Blockchain Hackathon 2018
- University Innovation Fellow, Stanford University
- Google Developer Student Clubs Lead